

Mathematical Statistics I
HOMEWORK 6

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1.

a.

$$\int_0^1 ky(1-y)dy = 1$$
$$\int_0^1 y(1-y)dy = \int_0^1 (y - y^2)dy = \left[\frac{y^2}{2} - \frac{y^3}{3} \right]_0^1 = \frac{1}{2} - \frac{1}{3} = \frac{1}{6}$$
$$\frac{1}{6}k = 1$$
$$k = 6$$

b.

$$P(0.4 \leq Y \leq 1) = \int_{0.4}^1 6y(1-y)dy$$
$$P(0.4 \leq Y \leq 1) = \int_{0.4}^1 6(1-y)dy = 6 \left[\frac{y^2}{2} - \frac{y^3}{3} \right]_{0.4}^1 = 6 \left[\left(\frac{1}{2} - \frac{1}{3} \right) - \left(\frac{0.16}{2} - \frac{0.064}{3} \right) \right]$$
$$P(0.4 \leq Y \leq 1) = 6 \left[\frac{1}{6} - (0.08 - 0.02133) \right] = 0.648$$
$$P(0.4 \leq Y \leq 1) = 0.648$$

c.

For a continuous variable, the probability at a single point is 0.

$$P(Y = 1) = 0$$

$$P(0.4 \leq Y < 1) = P(0.4 \leq Y \leq 1) = \int_{0.4}^1 f(y)dy$$

$$f(y) = 6y(1-y)$$

$$P(0.4 \leq Y < 1) = 0.648$$

d.

$$P(Y \leq 0.4 | Y \leq 0.8) = \frac{\int_0^{0.4} 6y(1-y)dy}{\int_0^{0.8} 6y(1-y)dy} = \frac{[3y^2 - 2y^3]_0^{0.4}}{[3y^2 - 2y^3]_0^{0.8}} = \frac{0.48 - 0.128}{1.92 - 1.024}$$
$$P(Y \leq 0.4 | Y \leq 0.8) = \frac{0.352}{0.896} = \frac{11}{28}$$
$$P(Y \leq 0.4 | Y \leq 0.8) = \frac{11}{28}$$

2.

Given,

$$f(y) = 6y(1 - y), \quad 0 \leq y \leq 1$$
$$P(Y \leq m) = 0.5$$

$$\int_0^m f(y) dy = 0.5$$

$$0.5 = \int_0^m 6y(1 - y) dy$$

$$0.5 = 6 \int_0^m (y - y^2) dy$$

$$0.5 = 6 \left[\frac{y^2}{2} - \frac{y^3}{3} \right]_0^m$$

$$0.5 = 6 \left(\frac{m^2}{2} - \frac{m^3}{3} \right)$$

$$0.5 = 3m^2 - 2m^3$$

$$2m^3 - 3m^2 + 0.5 = 0$$

$$2(0.5)^3 - 3(0.5)^2 + 0.5$$

$$(0.5)^3 = 0.125, \quad (0.5)^2 = 0.25$$

$$2(0.125)^3 - 3(0.25)^2 + 0.5 = 0.25 - 0.75 + 0.5 = 0$$
$$0 = 0$$

Therefore,

$$m = 0.5$$

3.

a.

$$F'(y) = \frac{d}{dy} (1 - e^{-y^2})$$

$$F'(y) = 2ye^{-y^2}$$

$$y \geq 0, 2ye^{-y^2} \geq 0$$

$F(y)$ never decreases

$$\lim_{y \rightarrow \infty} F(y) = 1 - e^0 = 0$$

$$\lim_{y \rightarrow \infty} F(y) = 1 - e^{-\infty} = 1$$

The function starts at 0 and approaches 1. These are properties of a valid cumulative distribution function.

b.

$$\begin{aligned}
 F(y) &= 1 - e^{-y^2} \\
 1 - e^{-y^2} &= 0.30 \\
 e^{-y^2} &= 0.70 \\
 -y^2 &= \ln(0.70) \\
 y^2 &= -\ln(0.70) \\
 y_{0.3} &= \sqrt{-\ln(0.70)} \\
 y_{0.3} &= \sqrt{0.3567} = 0.597 \\
 \text{The 30}^{\text{th}} \text{ percentile is } 0.597
 \end{aligned}$$

c.

$$\begin{aligned}
 f(y) &= \frac{d}{dy} F(y) \\
 f(y) &= \frac{d}{dy} (1 - e^{-y^2}) \\
 f(y) &= 2ye^{-y^2}, \quad y \geq 0
 \end{aligned}$$

d.

$$\begin{aligned}
 P(Y \geq 2) &= 1 - P(Y < 2) \\
 P(Y \geq 2) &= 1 - F(2) \\
 P(Y \geq 2) &= 1 - [1 - e^{-2^2}] = e^{-4} \\
 P(Y \geq 2) &\sim 0.0183
 \end{aligned}$$

The probability that the transistor operates for at least 2 (hundred) hours is about 0.0183.

4.

a.

$$\begin{aligned}
 E(Y) &= \int_0^4 yf(y)dy \\
 E(Y) &= \int_0^4 y \left(\frac{3}{64} y^2 (4 - y) \right) dy \\
 E(Y) &= \frac{3}{64} \int_0^4 y^3 (4 - y) dy \\
 E(Y) &= \frac{3}{64} \int_0^4 (4y^3 - y^4) dy \\
 E(Y) &= \frac{3}{64} \left[y^4 - \frac{y^5}{5} \right]_0^4 \\
 &\quad y = 4 \\
 y^4 - \frac{y^5}{5} &= (4^4) - \left(\frac{(4^5)}{5} \right) = 256 - \frac{1024}{5} = \frac{256}{5} \\
 E(Y) &= \frac{3}{64} \cdot \frac{256}{5} = \frac{768}{320} = 2.4 \\
 E(Y) &= 2.4
 \end{aligned}$$

The expected value of weekly CPU time is 2.4 hours.

$$\begin{aligned}
 E(Y^2) &= \int_0^4 y^2 f(y) dy \\
 E(Y^2) &= \frac{3}{64} \int_0^4 y^4 (4 - y) dy \\
 E(Y^2) &= \frac{3}{64} \int_0^4 (4y^4 - y^5) dy \\
 E(Y^2) &= \frac{3}{64} \left[\frac{4y^5}{5} - \frac{y^6}{6} \right]_0^4 \\
 &\quad y = 4
 \end{aligned}$$

$$\begin{aligned}
 E(Y^2) &= \frac{4(4^5)}{5} - \frac{(4^6)}{6} = \frac{4096}{5} - \frac{4096}{6} = \frac{2048}{15} \\
 E(Y^2) &= \frac{3}{64} \cdot \frac{2048}{15} = \frac{6144}{960} = 6.4 \\
 E(Y^2) &= 6.4 \\
 V(Y) &= E(Y^2) - [E(Y)]^2 \\
 V(Y) &= 6.4 - (2.4)^2 = 0.64 \\
 V(Y) &= 0.64
 \end{aligned}$$

The variance of weekly CPU time is 0.64.

b.

$$\begin{aligned}
 C &= 200Y \\
 a &= 200 \\
 E(Y) &= 2.4 \\
 V(Y) &= 0.64
 \end{aligned}$$

$$\begin{aligned}
 E(C) &= aE(Y) \\
 E(C) &= 200 \cdot 2.4 = 480 \\
 V(C) &= a^2 V(Y) \\
 V(C) &= (200)^2 \cdot 0.64 = 25,600
 \end{aligned}$$

The expected value of the weekly cost for the CPU time is \$480.

The variance of the weekly cost for CPU time is 25,600.

5.

a.

$$\begin{aligned}
 Y &\sim \text{Uniform}(1, 5) \\
 \theta_1 &= 1 \\
 \theta_2 &= 5 \\
 F(x) &= \frac{x - 1}{5 - 1} = \frac{x - 1}{4}, \quad 1 \leq x \leq 5
 \end{aligned}$$

$$P(Y > 2) = 1 - F(2) = 1 - \frac{2-1}{4} = 1 - \frac{1}{4} = 0.75$$

$$P(Y > 2) = 0.75$$

b.

$$Y \sim \text{Uniform}(1,5)$$

$$a = 1, b = 5$$

$$E[C] = E[c_0 + c_1 Y^2] = c_0 + c_1 E[Y^2]$$

$$E(Y^2) = \frac{a^2 + ab + b^2}{3}$$

$$E(Y^2) = \frac{(1^2) + (1)(5) + (5^2)}{3} = \frac{31}{3}$$

$$E[C] = c_0 + \frac{31}{3} c_1$$

6.

$$Y \sim \text{Uniform}(0,500)$$

$$f(y) = \frac{1}{500 - 0} = \frac{1}{500} \quad , \quad 0 \leq y \leq 500$$

a.

$$475 \leq Y \leq 500$$

$$P(475 \leq Y \leq 500)$$

$$P(475 \leq Y \leq 500) = \int_{475}^{500} \frac{1}{500} dy = \frac{1}{500} [y]_{475}^{500} = \frac{1}{500} (500 - 475)$$

$$P(475 \leq Y \leq 500) = \frac{25}{500} = 0.05$$

$$P(Y \text{ within } 25 \text{ ft of the end}) = 0.05$$

b.

$$0 \leq Y \leq 25$$

$$P(0 \leq Y \leq 25)$$

$$P(0 \leq Y \leq 25) = \int_0^{25} \frac{1}{500} dy = \frac{1}{500} [y]_0^{25} = \frac{1}{500} (25 - 0)$$

$$P(0 \leq Y \leq 25) = \frac{25}{500} = 0.05$$

$$P(Y \text{ within 25ft of the beginning}) = 0.05$$

c.

$$distance_{beg} < distance_{end}$$

$$Y < 500 - Y$$

$$2Y < 500$$

$$Y < 250$$

$$Y \sim \text{Uniform}(0, 500)$$

$$F(x) = \frac{x - 0}{500 - 0} = \frac{x}{500}$$

$$P(Y < 250) = F(250) = \frac{(250)}{500} = 0.5$$

$$P(\text{closer to beginning than end of line}) = 0.50$$